

**AP<sup>®</sup> STATISTICS**  
**2010 SCORING GUIDELINES (Form B)**

**Question 1**

**Intent of Question**

The primary goals of this question were to assess students' ability to (1) compare three distributions of a quantitative variable; (2) construct a stemplot; (3) recognize that different graphical displays of the same data can reveal different characteristics of a distribution.

**Solution**

**Part (a):**

Comparing the medians reveals that the concentration of aldrin tends to be highest for River X and lowest for River Z. About 50 percent of the concentrations of aldrin for Rivers X and Y are higher than all of the concentrations for River Z. River X also displays the most variability in aldrin concentrations, as seen by the largest range and largest IQR, and River Z has the least variability, as judged by both IQR and range. The shapes of the three distributions differ, in that the distribution appears to be skewed to the right for River X, roughly symmetric for River Y and slightly skewed to the left for River Z.

**Part (b):**

Aldrin concentrations (in ppm) for River X  
Leaf unit = 0.1 (for example, 3 | 4 represents 3.4 ppm)

|   |  |         |
|---|--|---------|
| 3 |  | 47      |
| 4 |  | 0236678 |
| 5 |  | 13356   |
| 6 |  |         |
| 7 |  | 35      |
| 8 |  | 0267    |

**Part (c):**

The stemplot shows a gap in the distribution of aldrin concentrations for River X between the values of 5.6 and 7.3 ppm of aldrin. This gap is not apparent in the boxplot.

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**Question 1**

**Intent of Question**

The primary goals of this question were to assess students' ability to (1) describe and use a procedure for estimating medians from histograms; (2) use graphical displays to compare two different distributions; (3) use graphical and numerical information to compare the means for two groups.

**Solution**

**Part (a):**

The median is the value with half of the P-T ratios at or below it and half of the values at or above it.

For  $n$  observations in a group, use  $\frac{n+1}{2}$  to find the position of the median in the ordered list of observations.

For states west of the Mississippi ( $n = 24$ ) the median falls between the 12th and 13th value in the ordered list, and both the 12th and 13th values fall in the interval 15–16. For states east of the Mississippi ( $n = 26$ ) the median falls between the 13th and 14th value in the ordered list, and both of these values also fall in the interval 15–16.

From the histogram, cumulative frequencies for the two groups are shown in the table below.

| Interval | West                 | East                  |
|----------|----------------------|-----------------------|
| 12–13    | 1                    | 2                     |
| 13–14    | $1 + 4 = 5$          | $2 + 4 = 6$           |
| 14–15    | $1 + 4 + 6 = 11$     | $2 + 4 + 4 = 10$      |
| 15–16    | $1 + 4 + 6 + 3 = 14$ | $2 + 4 + 4 + 11 = 21$ |

Thus, the median P-T ratio for both groups is at least 15 students per teacher and at most 16 students per teacher.

**Part (b):**

The shapes of the two histograms are different. The histogram for states that are west of the Mississippi River is unimodal and skewed to the right, whereas the histogram for states that are east of the Mississippi River is unimodal and nearly symmetric.

As noted in part (a), the medians of the two distributions are about the same, between 15 and 16 for both distributions.

The histograms also show that there is more variability in the P-T ratios for states that are west of the Mississippi River. Although the greatest and least values for each group are not known, the range can be approximated. The range for the west is at most  $22 - 12 = 10$ , and the range for the east is at most  $19 - 12 = 7$ .

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**Question 1 (continued)**

**Part (c):**

The medians of the two distributions are about the same, as determined in part (a). The distribution of P-T ratios for states that are west of the Mississippi River is skewed to the right, indicating that the mean will probably be higher than the median. The rough symmetry for the east group indicates that the mean will be close to the median. Thus, the mean for the west group will probably be greater than the mean for the east group.

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**Question 2**

**Intent of Question**

The primary goals of this question were to assess students' ability to (1) describe a simple random sampling procedure; (2) identify an effective stratification variable; (3) provide a statistical advantage of a stratified random sample over a simple random sample in context.

**Solution**

**Part (a):**

The administrators could number an alphabetical list of students from 1 to 2,500. They could then use a random number generator from a calculator or computer to generate 200 unique random integers from 1 to 2,500. The students corresponding to those 200 numbers would be asked to participate in the survey.

**Part (b):**

One possible stratification variable might be the school level of the student (elementary, middle, high school). The students' perceptions of the importance of good nutrition in food served may differ depending on the students' ages and therefore on school levels. For example, there may be a difference between what elementary students value in food served as opposed to middle school and high school students.

**Part (c):**

One statistical advantage of using stratified random sampling as opposed to simple random sampling is, for example, if the elementary, middle and high school strata create groups that differ with respect to what they value — and are therefore more homogeneous with respect to opinion on this issue — then for the same overall sample size a more accurate estimate of the overall proportion of students who are satisfied with the food under this contract may result. Another advantage is that stratified random sampling guarantees that each of the school-level strata will have some representation, because it is possible that a simple random sample would miss one or more of the strata completely.